MUTHU BALAMURUGAN

ARTIFICIAL INTELLIGENCE & DATA SCIENCE ENGINEER

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1. PROFESSIONAL SUMMARY

Innovative Artificial Intelligence & Data Science student at Ramco Institute of Technology.
Focused on engineering real-world software across generative AI pipelines, computer vision, and industrial IoT edge telemetry with high performance and resilient system architecture.

2. EDUCATION & ACADEMIC CREDENTIALS

Bachelor of Technology - Artificial Intelligence & Data Science

Ramco Institute of Technology | Timeline: 2025 - 2029

Cumulative CGPA: 7.8 / 10.0

Key Coursework: Machine Learning, Neural Networks, Computer Vision, Database Systems, DSA

3. TECHNICAL SKILLS MATRIX

Programming Languages: Python, TypeScript, JavaScript, C++, Java, SQL

AI / ML & Computer Vision: PyTorch, OpenCV, Scikit-Learn, Google Gemini API, MediaPipe

Web & Distributed Systems: React.js, Next.js, FastAPI, Node.js, Express, Supabase, PostgreSQL

IoT, Firmware & Tools: Arduino, ESP32, MQTT Protocol, Git/GitHub, Docker, Linux

4. HIGH-IMPACT ENGINEERING PROJECTS (PART 1)

EchoVerse AI - Temporal Multiverse & Scenario Engine (2025)

`Tech Stack: React, FastAPI, Python, Google Gemini 2.0 API, Server-Sent Events`

- Architected generative timeline engine enabling branching narrative and alternative reality simulations.
- Engineered low-latency streaming pipeline delivering sub-200ms token completion speeds.
- Designed graph-based decision tree visualization for instant rollback and scenario navigation.

AI Student OS - Integrated Academic Intelligence Workspace (2025)

`Tech Stack: Next.js 14, TypeScript, Supabase, PostgreSQL, TailwindCSS`

- Engineered comprehensive productivity suite with automated flashcards generation and task boards.
- Built real-time query velocity tracker and relational grade schema using PostgreSQL and Row Level Security.
- Designed responsive Neo-Brutalist dashboard delivering fluid mobile and desktop accessibility.

MUTHU BALAMURUGAN - CREDENTIALS & EXPERIENCE DOSSIER

Hardware Systems, Certified Industry Competencies, Job Simulations & Collegiate Honors

5. HIGH-IMPACT ENGINEERING PROJECTS (PART 2 - HARDWARE & VISION)

Industrial Motor Health Monitor - Non-Invasive Edge Telemetry (2024)

Tech Stack: Arduino, Embedded C++, Python, MQTT Protocol, Accelerometer, Thermistor

- Designed IoT condition-monitoring rig sampling 3-axis vibration FFT spectrum and real-time temperatures.
- Automated MQTT alert triggers for bearing anomaly detection, mitigating equipment downtime.

Eye Moment - Hands-Free Assistive Gaze Cursor Controller (2024)

Tech Stack: Python, OpenCV, MediaPipe, PyAutoGUI, NumPy

- Engineered accessibility interface tracking 468 3D facial mesh points for responsive gaze cursor steering.
- Achieved 60 FPS real-time responsiveness with ergonomic blink-detection gesture controls.

6. VERIFIED INDUSTRY CERTIFICATIONS

1. **Data Science & Analytics** - HP (2024)
   Skills: Data Analytics, Statistical Analysis, Data Visualization, Predictive Modeling
2. **Introduction to Cybersecurity Awareness** - HP (2024)
   Skills: Cybersecurity Fundamentals, Threat Awareness, Data Protection, Network Safety
3. **Cyber Job Simulation** - Forage (2024)
   Skills: Incident Response, Vulnerability Assessment, Security Log Analysis, Defense Simulation
4. **Controllers Job Simulation** - Forage (2024)
   Skills: Financial Governance, Risk Assessment, Process Automation, Internal Control Auditing
5. **Explore Machine Learning using Python** - Infosys (2024)
   Skills: Machine Learning Pipelines, Scikit-Learn, Feature Engineering, Model Evaluation

7. PRACTICAL INDUSTRY SIMULATIONS & INTERNSHIPS

Forage Cybersecurity Virtual Experience (2024):
Completed enterprise threat defense scenarios, incident response workflows, and log triage protocols.

Forage Financial Controllership Simulation (2024):
Executed risk evaluations, automated financial reconciliation checks, and internal audit compliance.

8. COMPETITIVE HACKATHONS & ACHIEVEMENTS

2x Collegiate Hackathon Finalist (2024 - 2025):
Engineered rapid-turnaround generative AI and hardware prototypes under strict 24-hr sprint limits.

Technical Symposium Exhibitor (2024):
Presented live demonstrations of hands-free computer vision assistive tools and industrial edge sensing.